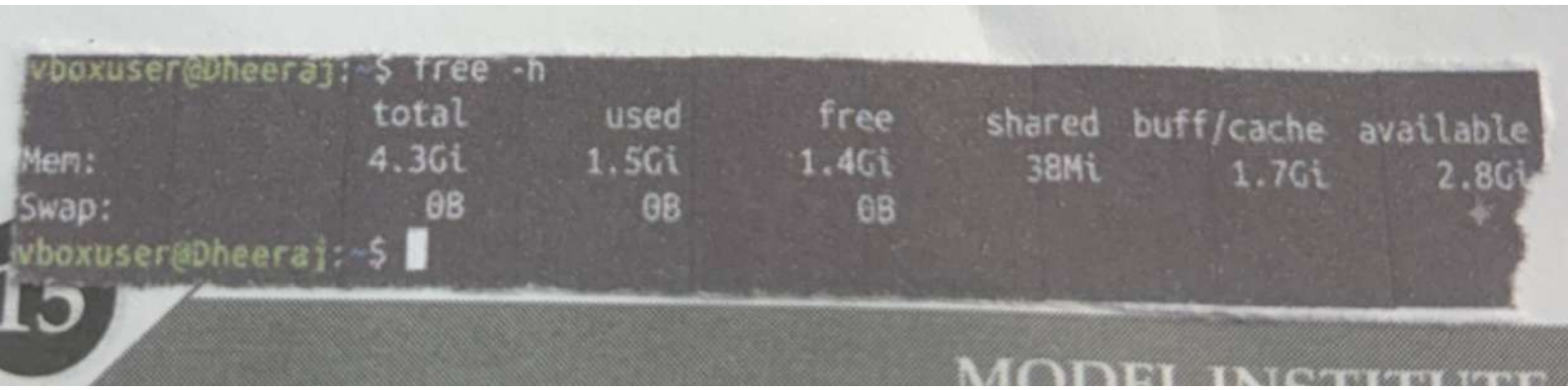
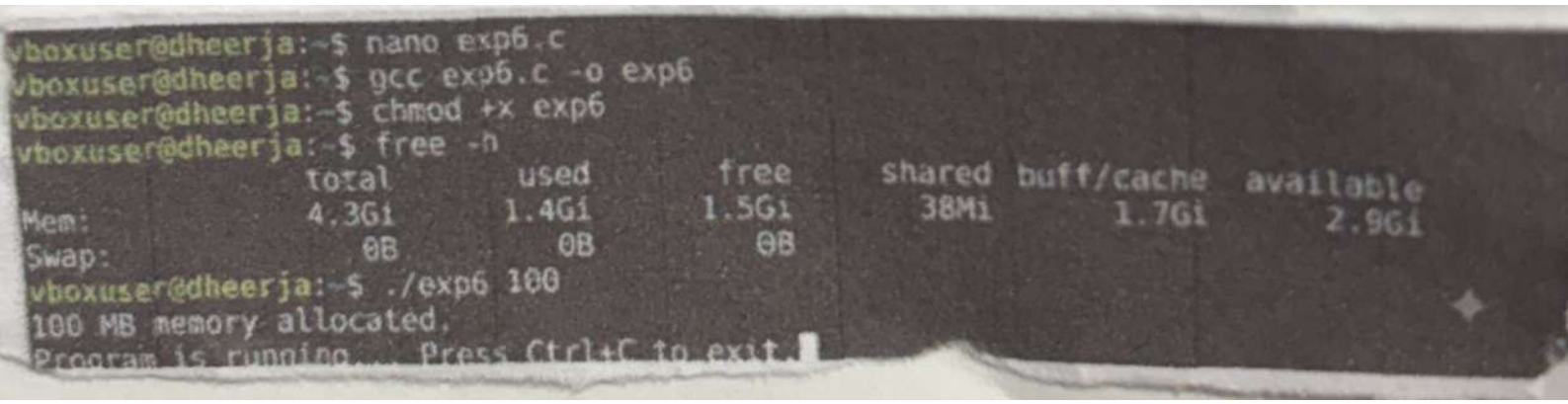

vboxuser@Dheeraj:~$ free -h
total used free shared buff/cache available
Mem: 4.3Gi 1.5Gi 1.4Gi 38Mi 1.7Gi 2.8Gi
Swap: 0B 0B 0B
vboxuser@Dheeraj:~$


vboxuser@dheerja:~$ nano exp6.c
vboxuser@dheerja:~$ gcc exp6.c -o exp6
vboxuser@dheerja:~$ chmod +x exp6
vboxuser@dheerja:~$ free -h
total used free shared buff/cache available
Mem: 4.3Gi 1.4Gi 1.5Gi 38Mi 1.7Gi 2.9Gi
Swap: 0B 0B 0B
vboxuser@dheerja:~$ ./exp6 100
100 MB memory allocated.
Program is running... Press Ctrl+C to exit.

GNU nano 8.7.1

```c
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>

int main(int argc,char *argv[])
{
        int mb;

        if(argc != 2)
        {
           printf("Usage: %s <memory_in_MB>\n", argv[0]);
           return 1;
        }
        mb = atoi(argv[1]);

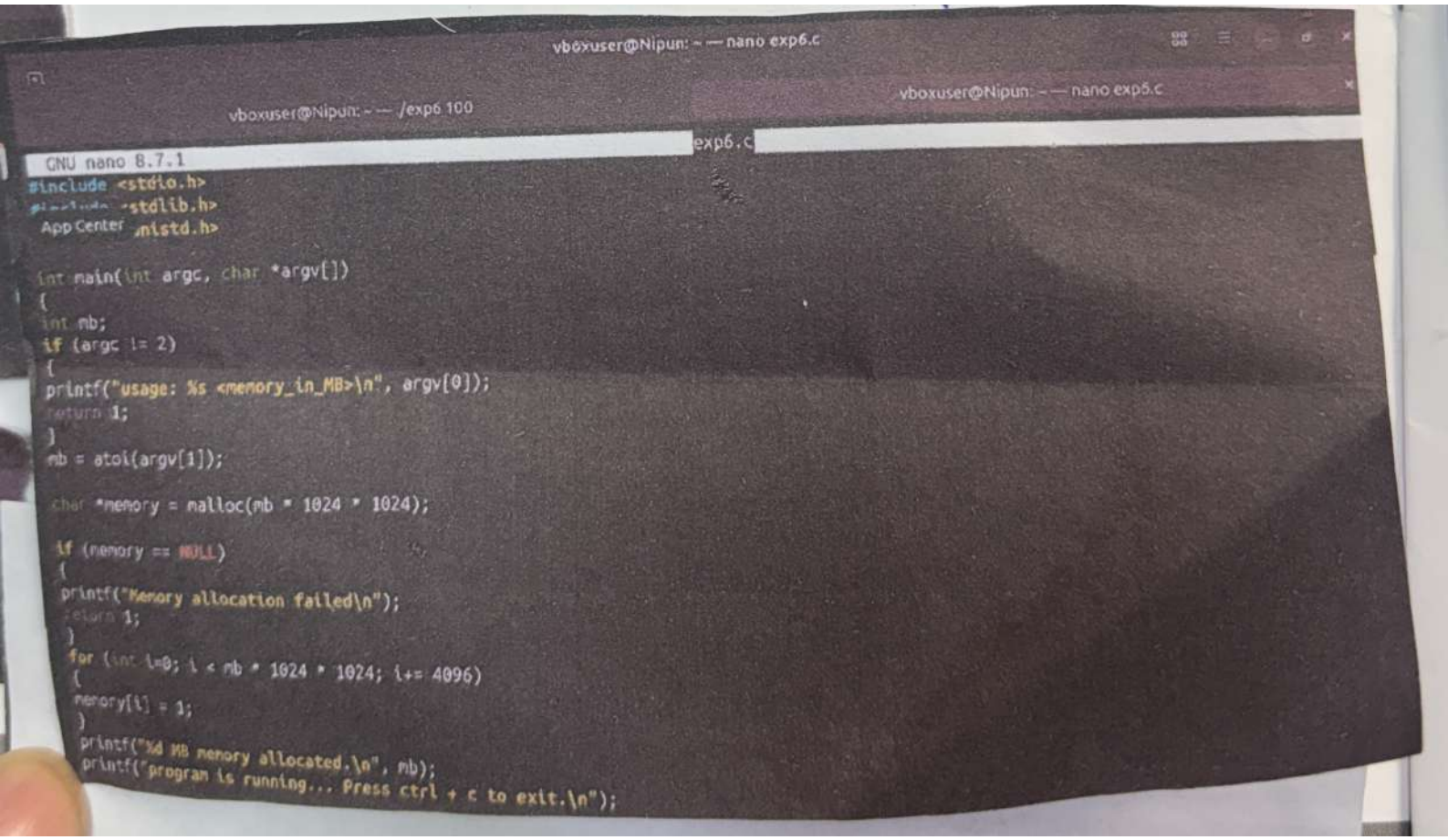
        char *memory = malloc(mb * 1024 * 1024);

        if (memory == NULL)
        {
           printf("Memory allocation failed\n");
           return 1;
        }

        for(int i=0; i < mb * 1024 * 1024; i += 4096)
        {
            memory[i] = 1;
        }


        printf("%d MB memory allocated.\n", mb);
        printf("Program is running... Press Ctrl+C to exit.\n");

        while(1)
        {
            sleep(1);
        }
        return 0;
```


vboxuser@Nipun: ~ — nano exp6.c
vboxuser@Nipun: ~ — /exp6 100
vboxuser@Nipun: ~ — nano exp5.c
GNU nano 8.7.1
exp6.c
App Center
#include <stdio.h>
-stdlib.h>
.nistd.h>
int main(int argc, char *argv[])
{
int mb;
if (argc != 2)
{
printf("usage: %s <memory_in_MB>\n", argv[0]);
return 1;
}
mb = atoi(argv[1]);
char *memory = malloc(mb * 1024 * 1024);
if (memory == NULL)
{
printf("Memory allocation failed\n");
return 1;
}
for (int i=0; i < mb * 1024 * 1024; i+= 4096)
{
memory[i] = 1;
}
printf("%d MB memory allocated.\n", mb);
printf("progran is running... Press ctrl + c to exit.\n");